DAVID KURNIAWAN

davidk.academic@gmail.com | +62 812-2970-0811 | Sepang, Malaysia | GitHub | Portfolio | LinkedIn

Education

Xiamen University Malaysia Expected Sep 2028
B.Eng. (Hons.) in Artificial Intelligence GPA: 3.84/4.00
Dean's List: 3 consecutive semesters; Top 16% of cohort
Relevant Coursework: Data Structures, Object-Oriented Programming (C++), Applied Machine Learning, Deep Learning, Natural Language Processing

Technical Skills

Programming: Python, JavaScript, TypeScript, C, C++, HTML, CSS
Backend & Frontend: FastAPI, Pydantic, Flask, Node.js, Next.js, React, REST APIs, OpenAPI, WebSockets
Databases: PostgreSQL, Supabase, pgvector, Firebase, Qdrant
Cloud & Infrastructure: Docker, Cloudflare Workers, Railway, Vercel, Cloudflare DNS, Resend, CI/CD (GitHub Actions)
Engineering Practices: System design, authentication SQL migrations, pytest, unit and integration testing, logging, error handling, model evaluation
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, OpenCV, LangChain, LangGraph, Transformers (Hugging Face), FAISS, BM25, Weights & Biases
Languages: Indonesian (Native), English (IELTS 7.0), Mandarin Chinese (Elementary)

Selected Projects

Adept — AI-Powered Upskilling and Reskilling Platform 2026
Full-Stack and AI Engineering / Python, TypeScript, FastAPI, Next.js, React, PostgreSQL, pgvector, Pydantic, Docker, GitHub Actions

- Led a four-person team building a personalized upskilling and reskilling platform; engineered learning pathways across 98 sourced courses using skills-gap analysis, prerequisite ordering, and time and budget constraints.
- Integrated a data-grounded large language model (LLM) assistant with structured outputs, catalogue-ID validation,PII redaction and provider failover; implemented deterministic fallback responses in English, Malay and Chinese.
- Implemented semantic skill suggestions using Gemini embeddings, pgvector storage and cosine similarity; integrated 1,210 job postings from 127 employers to connect learning pathways with job-specific skill requirements.
- Strengthened backend reliability through PostgreSQL connection pooling, caching and bounded concurrency; automated pytest, database integration tests, OpenAPI contract checks and production builds using GitHub Actions CI/CD.

KerjaCerdas — AI Job-Matching Platform 2026
Lead Engineer / Backend / FastAPI, LangGraph, PostgreSQL/pgvector, Docker, React

- Built an AI job-matching platform using FastAPI, LangGraph, PostgreSQL and React; implemented GitHub Actions CI, vector-search integration tests, Alembic migration checks, and CV PII redaction before LLM inference.
- Engineered explainable candidate ranking using 768-dimensional Gemini embeddings, pgvector HNSW semantic search and weighted matching scores; distinguished self-reported, quiz-verified and employer confirmed skills to make candidate evidence explicit.

Indonesian Legal RAG 2026
LangChain, FAISS, BM25, Hugging Face, Weights & Biases

- Built a retrieval-augmented generation (RAG) pipeline combining BM25, FAISS dense retrieval, HyDE query expansion and cross-encoder reranking, with web-search fallback for low-confidence queries.
- Fine-tuned Llama 3 8B using 4-bit QLoRA for Indonesian legal instruction following; tracked experiments with Weights & Biases and published the model to Hugging Face.

Awards & Achievements

Top 41 / 2,000+ Teams — PIDI × Digdaya Hackathon (Tier 3) Mar – Sep 2026
Led engineering for KerjaCerdas, overseeing system architecture, AI inference, and full-stack implementation.
Silver Award — SEA-CICSIC 2026 International Research Competition Jun 2026
Led AI strategy and developed the proposal and pitch deck for Omni-QC, an AI-based manufacturing defect-prediction and quality-control concept.
3rd Place — DPickleball AI Competition Oct 2025
Led a 3-person team developing a reinforcement-learning agent using RecurrentPPO, LSTM, CNN, and Unity ML-Agents; implemented reward shaping and self-play league training.
Top 20% Globally — International Quant Championship, Stage 1 Apr 2025